Journal of Hydrology 384 (2010) 33–45

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Cross-well slug interference tests: An effective characterization method for resolving aquifer heterogeneity

R. Brauchler [a,*], R. Hu [a], T. Vogt [a,1], D. Al-Halbouni [b], T. Heinrichs [a], T. Ptak [a], M. Sauter [a]

[a] Geoscience Centre, University of Göttingen, Goldschmidtstraße 3, 37077 Göttingen, Germany
[b] Geophysics Department, University of Göttingen, Friedrich-Hund-Platz 1, 37077 Göttingen, Germany

ARTICLE INFO

Article history:
Received 26 March 2009
Received in revised form 30 November 2009
Accepted 7 January 2010

This manuscript was handled by P. Baveye, Editor-in-Chief, with the assistance of D.P. Ahlfeld, Associate Editor

Keywords:
Highly permeable aquifers
Cross-well slug interference tests
Hydraulic tomography
Wellbore storage
Inversion theory

SUMMARY

In this study the potential of cross-well slug interference tests for high resolution aquifer characterization of hydraulic heterogeneity was assessed. The cross-well slug interference tests were performed at the research site "Stegemühle", located in the Leine River valley near Göttingen, Germany. The geological composition of the subsurface, consisting mainly of 3.5 m silt and clay overlying 2.5 m sand and gravel, was determined by geophysical well logging and bore core data. To account for lateral changes a refraction seismic survey was conducted. Based on these data an area, characterized by an aquifer thickness of approximately 2 m and an average hydraulic conductivity of 5.0×10^{-4} m/s (determined by pumping tests), most appropriate for cross-well slug interference tests, was chosen. Altogether 196 cross-well slug interference tests were performed using a tomographic measurement array. The cross-well slug interference tests were evaluated using type curve analysis, which provided detailed information concerning the vertical changes of hydraulic conductivity and specific storage. To assess hydraulic strata connectivity a travel time based tomographic inversion approach was utilized. The potential of the inversion approach to determine lateral changes could be successfully demonstrated by the reconstruction of the pinch out of a high diffusivity layer close to the bottom of the aquifer. The results demonstrate that the combined evaluation of cross-well slug interference tests based on type curve analysis and travel time inversion allows for the development of a detailed model about subsurface hydraulic heterogeneity.

© 2010 Elsevier B.V. All rights reserved.

Introduction

Slug tests have traditionally been utilized as a technique to determine the hydraulic conductivity (K) of an aquifer over a relatively small scale. A slug test essentially consists of measuring the recovery of head in a well after a near instantaneous change in head at the well. The most important advantageous of slug tests are the low costs, its simplicity, the short duration (at least in high permeability media) and that no water needs to be handled, which is a very important advantage at sites of suspected groundwater contamination. When a slug test is performed in a cross-well mode, however, it can provide considerably more information. Despite the common perception that a slug test only affects a small volume of the aquifer in the vicinity of the test well (e.g. Ferris and Knowles, 1963; Rovey and Cherkauer, 1995), response data with a reasonable signal to noise ratio can be collected at distances of over several hundred times the radius of the screen of the test well (e.g. Barker and Black, 1983; Sageev, 1986). The area affected by a slug test is strongly dependent on the dimensionless storage parameter. The smaller the dimensionless storage parameter the larger the response amplitude in the observation well and the larger the radius of influence. Barker and Black (1983), have shown that the aquifer parameters derived from the analysis of a slug test are spatially weighted averages of the continuously varying parameters over the whole aquifer, whereby the weights will clearly decline with increasing distance from the well.

A number of analytical and semi-analytical methods for the evaluation of cross-well slug interference tests based on type curve analysis for different types of aquifers and geologic media have been reported in the literature (e.g. Karasaki et al., 1988; Novakowski, 1989; Chu and Grader, 1991; Hyder et al., 1994; Butler, 1995; Spane, 1996; Butler and Zhan, 2004; Audouin and Bodin, 2007).

However, there are only a small number of field studies that have intensively applied the different methods for detailed site characterization. Field studies that have used cross-well test characterization of sedimentary unconfined aquifers are reported by Spane et al. (1996) and Belitz and Dripps (1999). Spane (1996) proposed an analytical approach which is based on the transformation of type curves developed for constant-rate pumping tests to type curves for cross-well slug interference tests. In

* Corresponding author. Tel.: +49 5513912864; fax: +49 551399379.
E-mail address: rbrauch@gwdg.de (R. Brauchler).
[1] Present address: Swiss Federal Institute of Aquatic Science & Technology (EAWAG), Department Water Resources and Drinking Water, Überlandstraße 133, 8600 Dübendorf, Switzerland.

0022-1694/$ - see front matter © 2010 Elsevier B.V. All rights reserved.
doi:10.1016/j.jhydrol.2010.01.004

Spane et al. (1996) the method is applied to an aquifer composed of clayey sands and gravels. These investigations demonstrate a high correspondence between pumping test responses and slug interference responses which suggests that the test conditions imposed by the performed cross-well slug interference tests are comparable to those existing during the early stages of pumping tests. Belitz and Dripps (1999) applied the semi-analytical solution developed by Hyder et al. (1994) and Butler (1995), to the evaluation of cross-well slug interference tests performed in unconsolidated silts and clays. These investigations showed that the additional evaluation of the pressure response in an observation well next to the pressure response in the test well yields information about wellbore skin and aquifer anisotropy and improves the calculation accuracy of specific storage.

Field studies in fractured rocks are reported in Novakowski (1989) and Audouin and Bodin (2007). Novakowski (1989) derived a semi-analytical solution for cross-well slug test accounting for wellbore storage effects at the observation well. This analytical method was applied in the investigation of hydraulic properties of a horizontally arranged fracture zone within an Ordovician-aged shale. A semi-analytical solution for the interpretation of cross-well slug interference tests in fractured media was developed by Audouin and Bodin (2007) and is based on the previous work of Barker (1988). This solution approach account for inertial effects at both the test and the observation wells as well as for the fractional flow dimension in the aquifer. The developed solution has been used successfully for the evaluation of cross-well slug interference tests performed in fractured and karstic limestones. Based on the comparison of hydraulic conductivity values estimated from cross-well slug interference tests with those interpreted from pumping tests, Audouin and Bodin (2007) concluded that cross-well slug interference tests are an important characterization method for identifying the hydrodynamic properties of karstic channels and fracture flow paths.

In this study the potential of cross-well slug test to characterize the hydraulic heterogeneity in highly permeable porous granular aquifers is assessed. The pressure disturbances recorded in the observation wells are evaluated by two methods: (a) type curve analysis and (b) the application of a travel time based tomographic inversion approach, which is based on the analysis method presented in Brauchler et al. (2007). The data base comprises a suite of tomographic cross-well slug interference tests recorded between five wells which are arranged in a five-point star configuration. The center well of the five-point star configuration served as a test well, and the four surrounding wells served as observation wells. Fig. 1 shows the tomographic configuration and map plan deployment of the five-well test system. Each of the four vertical profiles comprises seven slugged intervals and seven observation points isolated with double packer systems. Thus, each profile consists of 49 transient pressure curves. For the design and evaluation of the cross-well slug interference tests detailed data were collected about the structural composition and the hydraulic properties of the aquifer by means of surface and wireline geophysics, bore core data, pumping tests and multi-level slug tests.

(a) As a first step in the analysis, the cross-well slug interference tests were analyzed using the (semi) analytical solution developed by Hyder et al. (1994) and Butler and Zhan (2004). This initial evaluation provided detailed information concerning the subsurface hydraulic properties. The usage of a double packer system allowed to determine hydraulic conductivity and specific storage for different depth intervals of the aquifer which provided first insight into aquifer heterogeneity in vertical direction. However lateral changes, e.g. the pinching out of a high permeable zone, cannot be identified by using existing analytical solutions.

(b) For the assessment of hydraulic strata connectivity a hydraulic travel time based tomographic approach was applied. This inversion scheme follows the procedure of seismic ray tomography and is based on the transformation of the transient ground water flow equation into the eikonal equation using an asymptotic approach (Virieux et al., 1994). The eikonal equation can be solved with ray tracing techniques or particle tracking methods, which allow the calculation of pressure propagation along trajectories (Vasco and Datta-Gupta, 1999; Vasco et al., 1999, 2000; Vasco and Karasaki, 2006; Kulkarni et al., 2000; Datta-Gupta et al., 2001; Brauchler et al., 2003, 2007; He et al., 2006). The applied inversion approach is an extension of the analysis method presented by Brauchler et al. (2007). The main feature of this procedure is a travel time integral relating the square root of the peak travel time, assuming a Dirac point source at the origin, to the inverse square root of diffusivity. Although, the travel time of a pressure signal between two wells is primarily a function of the hydraulic diffusivity (D), the ratio of hydraulic conductivity (K) and specific storage (S_s), the inversion results improve strongly the significance of the K and S_s estimates based on type curve analysis with regard to hydraulic strata connectivity.

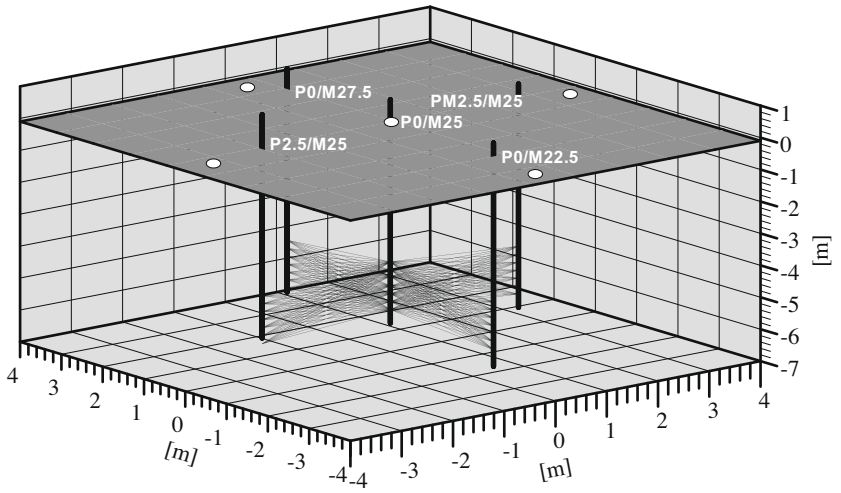


Fig. 1. Tomographic measurement five-star point set-up, illustrating the spatial position of the slugged intervals and observation points of four separate vertical response profiles. The white points indicate the position of the direct-push electrical conductivity logs.

Test site characterization

For assessing the applicability of cross-well slug interference tests for detailed hydrologic characterization, a dense, regularly arranged network consisting of wells with 0.0254 m and 0.0508 m inner diameters, was installed at the research site "Stegemühle", located in the Leine River valley near Göttingen.

For the well installations the direct-push technology was employed. The technology uses a hydraulic hammer supplemented with weight of the direct-push unit to push down drive rods to the desired depth of the projected well. The well casing consisting of high-density polyethylene (HDPE) pipes and slotted screens, is then lowered into the drive rods (inner diameter: 0.067 m, outer diameter 0.083 m). By retracting the drive rods, the formation is allowed to collapse back against the HDPE pipes. In total, a network of 16 wells was installed (Fig. 2).

Following well installation, site characterization was performed for the determination of the structural subsurface geometry, i.e. the definition of the base and the top of the aquifer, using next to geological logs, geophysical well logging information. Fig. 3 shows an example of the vertical profiles of the unconsolidated sediments of well P0/M22.5. The interpretation is based on an electrical conductivity log obtained from direct-push electrical probing, termed in the following DP-EC logs (e.g. Christy et al., 1994; Schulmeister et al., 2003), and a gamma-ray log. As shown in the geological interpretation, which was verified by subsurface material core analysis, the braided river sediments largely consist of 3.5 m silt and clay overlying 2.5 m sand and gravel. Bedrock consists of Triassic (Middle Keuper) mudstone. All recorded logs show the same basic pattern with variable thicknesses of the confining unit (silt and clay layers) and the aquifer material (sand and gravel). To examine these lateral heterogeneities, explained by the complexity of depositional and erosional processes in braided river systems (Huggenberger and Regli, 2006), a seismic refraction survey was conducted and the results were compared with those from the electrical conductivity logs. The spatial location of the refraction seismic profile is shown in Fig. 2. The data of the seismic profile were interpreted as a three layer case (Fig. 4):

- *Layer 1* exhibits an average velocity of 248 m/s, which is a typical value for soil.
- *Layer 2* exhibits an average velocity of 1596 m/s with an average depth of 2.18 m. The top boundary can be interpreted as a contact between saturated and unsaturated sediments. Note, the depth is not in agreement with water table measurements, but represents the capillary fringe. This is in agreement with the fact that at a water saturation of 90%, the wave velocity increases considerably (e.g. Sheriff and Geldart, 1995). The large difference between the aquifer top determined by DP-EC logs and the capillary fringe determined by refraction seismics can be explained by strong capillary forces which are typical for clay and silt confining units.
- *Layer 3* shows an average velocity of 2282 m/s with an average depth of 5.61 m. The layer is interpreted as mudstone. This mean, that the boundary between layers 2 and 3 corresponds to the aquifer bottom. The deviation of 0.5 m between the depth of the aquifer bottom determined with DP-EC logs and refraction seismics can be explained by the presence of a narrow alteration layer above the mudstone layer, which is in accordance with core samples. Fig. 4 shows that the boundary between layers 2 and 3 is characterized by a strongly variable relief. The construction of the aquifer bottom by a linear interpolation (indicated by white lines in Fig. 4) therefore, based on the DP-EC logs would be questionable. A sensitivity study based on wavefront inversion and ray tracing has shown an uncertainty of 0.5 m regarding the estima-

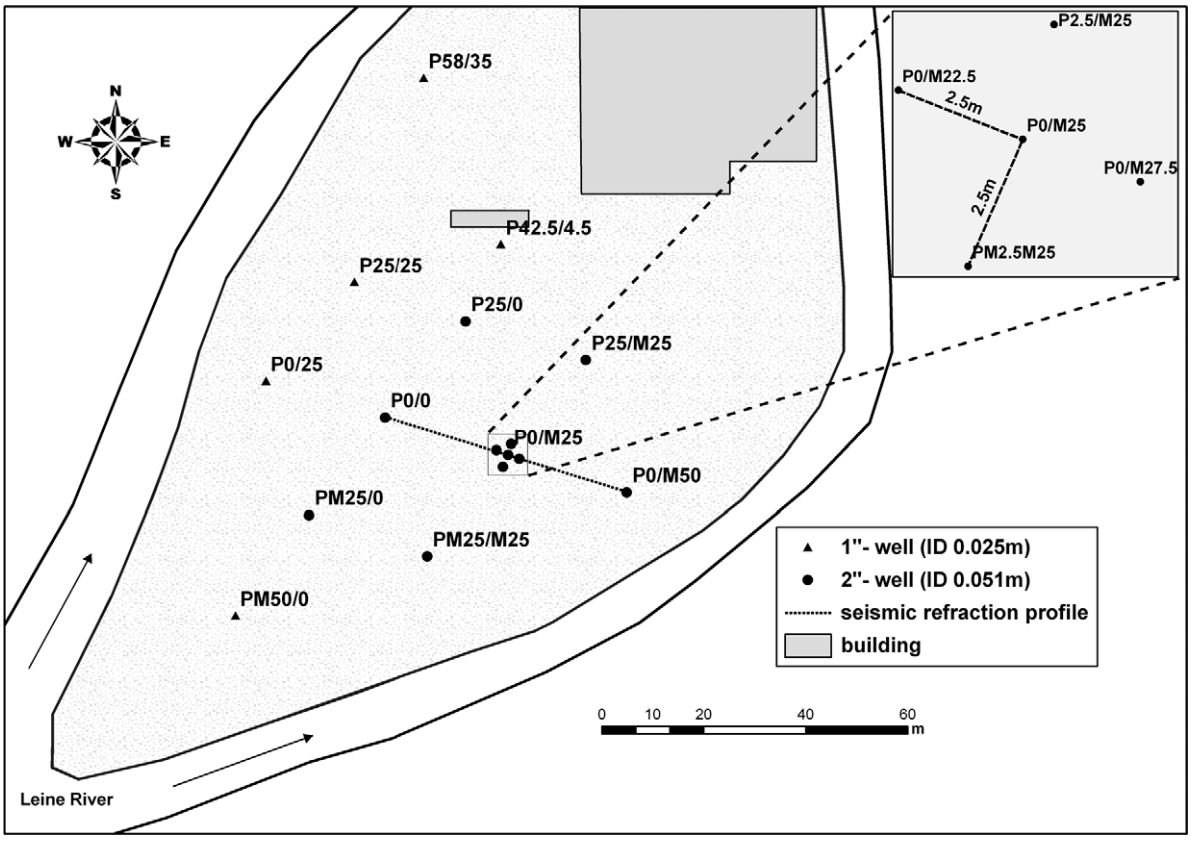


Fig. 2. Map of the installed well network and the seismic lines at the "Stegemühle" test site.

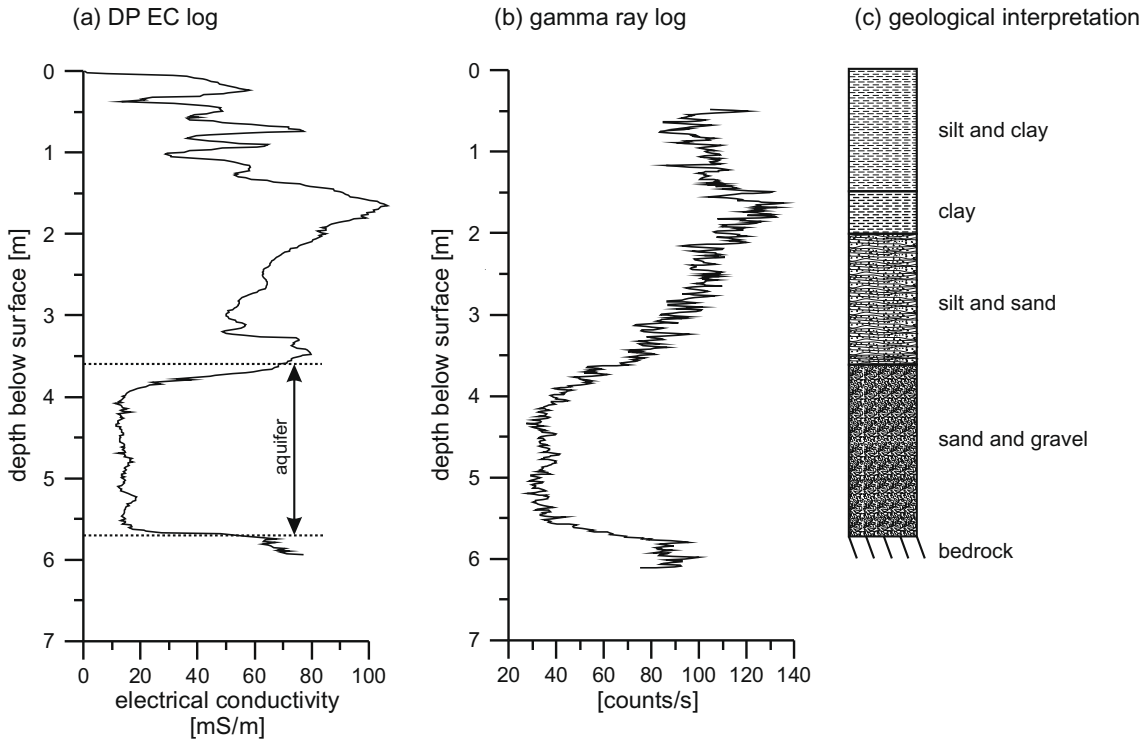


Fig. 3. Geological interpretation of the subsurface based on direct-push electrical conductivity logging and gamma-ray logging.

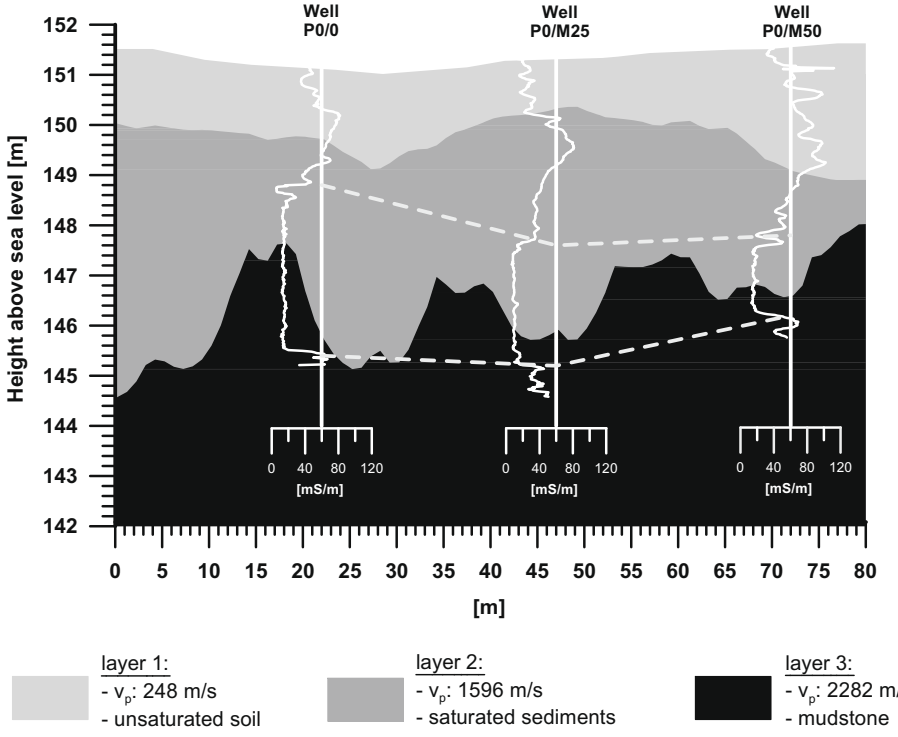


Fig. 4. Velocity model of the subsurface based on seismic refraction in comparison with direct-push electrical conductivity logs.

tion of the thickness of layer 3. However, the data indicate no uncertainty regarding the strongly variable relief of the mudstone layer.

Based on the detailed geometric information, obtained from the previous site characterization activities, an area was selected for the planned cross-well slug interference tests based on: (a) sufficient aquifer thickness and (b) adequate distance from aquifer boundaries in order to minimize the influence of changing boundary conditions. Applying these decision criteria an area was chosen, located in the vicinity of well P0/M25. In this area the aquifer thickness is approximately 2 m, and has an average hydraulic con-

ductivity of 5.0×10^{-4} m/s (determined by pumping tests). Four additional wells were installed at a distance of 2.5 m from the center well P0/M25. The spatial position of the five wells (in the following called five-point star) within the test site is illustrated in Fig. 2. For the high resolution structural characterization at the five-point star gamma-ray and DP-EC logging were performed at each well location. To visualize lateral changes in lithology, conditions were interpolated between the recorded well logs as illustrated in Fig. 5. The five gamma-ray and DP-EC logs show the same pattern as the logs recorded in well P0/M22.5 illustrated in Fig. 4. The high gamma-ray and EC values in the upper 3.5 m are interpreted as the confining unit consisting of silt and clay and the lower values in the interval 3.5–6 m indicate the spatial position of the confined aquifer material consisting of sand and gravel. The interpolation between the logs demonstrate that the lateral changes of the aquifer top and bottom do not exceed 0.5 m. Note, the DP-EC log were recorded at a distance of 1 m from the wells (indicated by white dots in Fig. 1).

Multi-level single-well slug tests were performed to determine the vertical changes of hydraulic conductivity within the aquifer. The slug tests were initiated pneumatically to avoid large fluctuations of the initial readings of the response signal (Butler, 1998). The head data were recorded with small diameter pressure transducers PDCR 35/D-8070) connected to a data logger (Campbell Scientific CR1000) with an acquisition rate of 10 Hz. The pressure transducer in the water column was placed close to the static water table to avoid the problem that transducer readings vary with installation depth and thus may not accurately measure the water-level position (Butler et al., 2003). Response data of the pneumatic slug test were normalized using the initial displacement measured by a pressure transducer installed in the air column. Fig. 6 shows the estimated hydraulic conductivity values versus depth for all wells of the five-point star. In each well, seven test depth intervals were investigated in order to characterize the entire aquifer thickness. The different test depth intervals were isolated using a double packer system. Four of the five profiles show that the lower 0.5 m section of the aquifer is characterized by a higher conductivity layer. Above this layer the values for hydraulic conductivity continuously decrease. The difference between the highest value at the bottom and the lowest values at the top of the aquifer is approximately one order of magnitude. These results are in accordance with the aquifer material core data which indicate that the upper part of the aquifer contains a larger silt component than the lower part. Only the hydraulic conductivity profile for well P0/M22.5 shows uniform hydraulic properties.

In Fig. 7a–c we have illustrated the response curves recorded in well P0/M27.5 for three different depths and the respective fits based on the analytical solution of Butler (1998). For the application of the solution the following assumption were made: Aquifer is confined, isotropic, nonbounded, the test well is partial penetrating and no well skin is present.

The response curve, recorded close to the bottom of the aquifer, shows an oscillatory/critically damped behavior, which is characteristic for an aquifer section of high hydraulic conductivity (Fig. 7a). The response curves recorded closer to the top of the aquifer exhibit overdamped, exponential decay responses, indicative of lower hydraulic conductivity materials. Following the field guidelines for slug tests in high permeable aquifers recommended by Butler et al. (2003), multi-stress level tests were performed using different initial displacement (0.1–0.2 m) to demonstrate a lack of stress dependence. As shown in Fig. 7d no stress dependence has exhibited.

Performance of cross-well slug interference tests

In this section the potential of cross-well slug interference tests to characterize the hydraulic heterogeneity in highly permeable porous aquifers is assessed. The cross-well slug interference tests were performed in a tomographic array, between five wells arranged as a five-point star. As noted previously, the center well of the five-point star configuration was used as test well, and the four surrounding wells served as observation wells. Fig. 1 shows the tomographic set-up. Each of the four illustrated profiles consists of seven slugged intervals and seven observation points isolated with double packer systems. Thus, each profile consists of 49 transient pressure curves.

A prerequisite for a successful evaluation of cross-well slug interference tests are data with a high temporal resolution. Since the sampling rate of the data logger depends on the number of pressure transducers used, the minimum number of three pressure transducers were used at the same time. The measurement set-up is illustrated in Fig. 8. Two pressure transducers were installed in

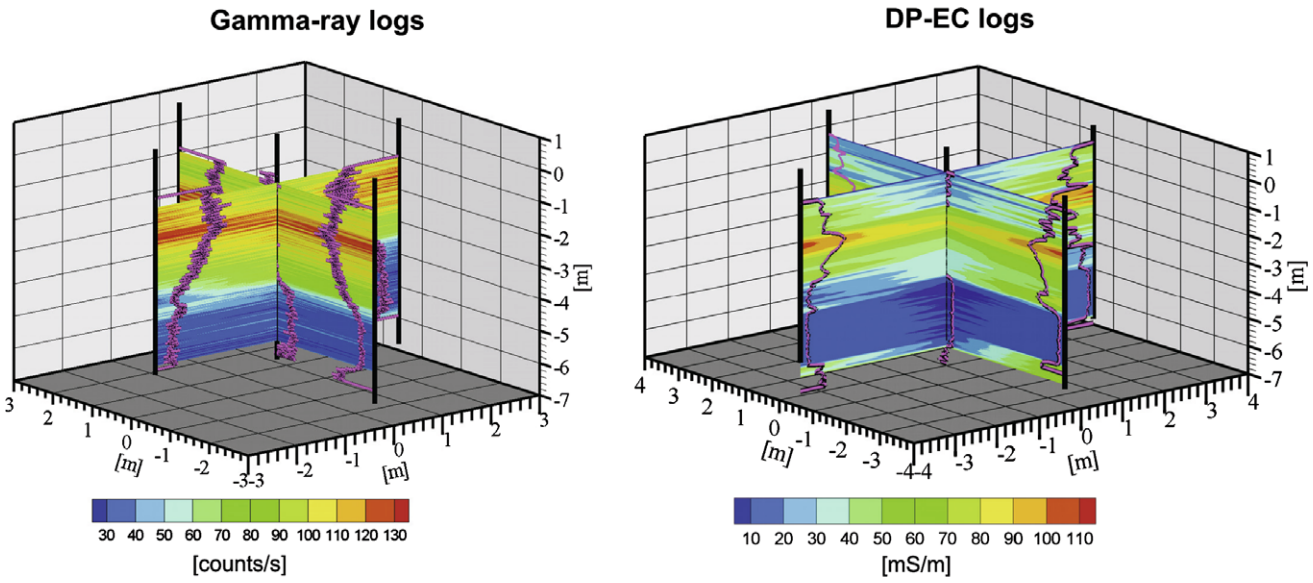


Fig. 5. Interpolated images of the gamma-ray and direct-push electrical conductivity logs. Note the distance between the EC-DP logs is 3.5 m in comparison to 2.5 m of the gamma-ray logs.

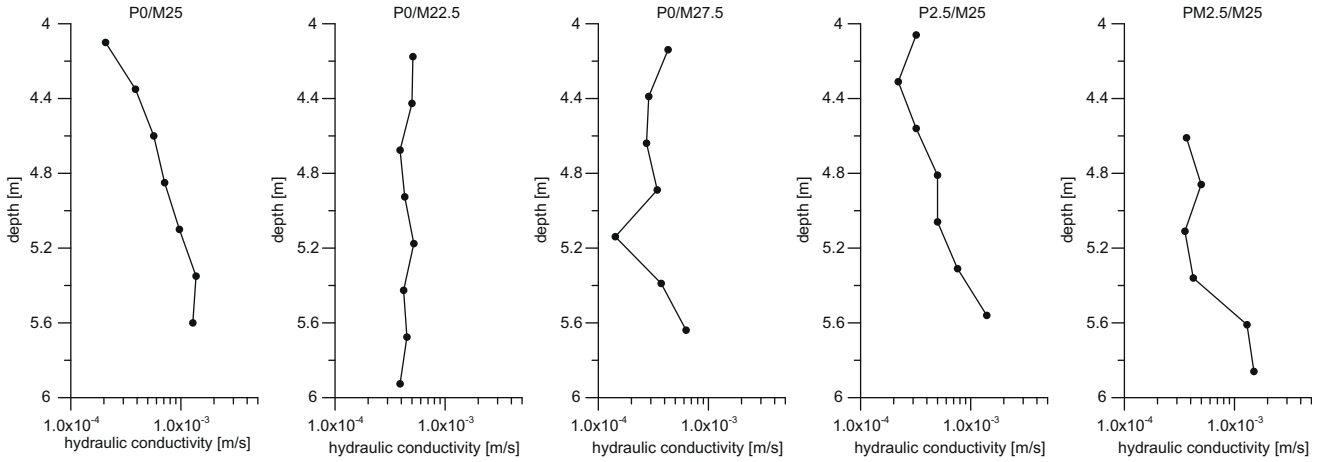


Fig. 6. Hydraulic conductivity estimates based on multi-level depth single-well slug testing.

the test well to measure the pressure drop in the air as well as in the water column of the test well. The third pressure transducer was installed in the screened section of the double packer system used in the observation well. Note, in both wells, in the test and in the observation well, a double packer system with an interval of 0.25 m and a ID of 0.031 m was employed. This measurement set-up allowed data to be recorded with a temporal resolution of 7 Hz. To ensure repeatability and data quality each slug test was conducted at least two times.

The cross-well slug interference tests were evaluated by type curve analysis with the objective of determining K and S_s at different depths across the whole aquifer thickness. For the assessment of strata connectivity, a tomographic travel time based inversion approach was applied, which enables the reconstruction of the hydraulic diffusivity distribution between wells. In a final step the K and S_s data obtained from the type curve analysis were used to improve the significance of the reconstructed diffusivity tomograms.

Evaluation based on type curve analysis

Fig. 9 shows the pressure responses in an observation well for three different depths. The tests were performed between well P0/M25 (test well) and well P0/M27.5 (observation well). Note the pressure response curves illustrated in Fig. 9 are recorded at the same position as the response curves from the multi-level single-well slug tests displayed in Fig. 7. For the evaluation (type curve matching) of the pressure response curves either the solution developed by (a) Butler and Zhan (2004) or (b) by Hyder et al. (1994) were applied. The solution developed by Butler and Zhan (2004) accounts for inertial effects in the test and observation wells and allows the analysis of underdamped and overdamped cross-well slug test data. This solution was applied to test data recorded in aquifer sections characterized by high hydraulic conductivity conditions, where inertial effects are evident and have to be taken into account. In Fig. 9b an example of a critically- or underdamped response curve (showing the characteristic oscillatory behavior) and the corresponding type curve are shown.

For the evaluation of pressure response curves recorded in aquifer sections exhibiting lower hydraulic conductivity conditions, the solution developed by Hyder et al. (2000), which neglects inertial effects, was applied. The pressure response curves are characterized by smaller and broader response amplitudes and exhibit no oscillatory behavior. In Fig. 9d the overdamped response curve and the corresponding analysis type curve are shown. For the application of the solution the following assumptions were made: Aquifer is confined, isotropic, nonbounded, the test well is partially penetrating and no well skin is at the test and observation wells present.

The illustrated pressure response curves are normalized to the initial displacement in the test well which varied between 0.2 m and 1 m in all performed cross-well slug interference tests. The observed maximum amplitude of the response curves recorded in the observation well varied between 0.005 and 0.04 m, indicating that the equipment has to offer reasonable signal to noise ratio characteristics to resolve very small amplitude response curves. For the performance of cross-well slug interference tests, larger initial displacements are needed, in comparison to single-well slug tests. In our case the recorded response curves in the test and observation wells show a dependence on the initial displacement, particularly in the lower part of the aquifer, which is characterized by high hydraulic conductivity values and exemplarily illustrated in Fig. 10. Note the test configuration in Fig. 10 is the same as in Fig. 9b. The stress dependence of the test and observation well responses indicates a nonlinear behavior and currently no analytical models are available, which completely account for these effects.

The dependence of the initial displacement of the test well responses (Fig. 10a) was not further investigated because the evaluation of these curves yield no information which goes beyond the evaluation of the multi-level single-well slug tests discussed in "Test site characterization". The evaluation of the observation well responses, however, allows to obtain bulk information about hydraulic conductivity and specific storage. These parameters cannot be assessed by multi-level single-well slug tests. Hence, in the following, the influence of these nonlinear effects on the estimation of K and S_s, derived from observation well responses, was assessed exemplarily by means of three observation well response curves illustrated in Fig. 10b. The initial displacements of the three cross-well slug interference tests, for which the response curves are illustrated, varied between 0.17 m and 0.51 m.

The hydraulic conductivity estimates derived from the observation well responses indicate no significant influence of nonlinear effects. For all three curves a hydraulic conductivity value of 0.0013 m/s was estimated, applying the analytical solution proposed by Butler and Zhan (2004). The specific storage estimates, however, show a variation of a factor of two and vary between 1.46×10^{-5} m^{-1} and 2.48×10^{-5} m^{-1}. To illustrate the strong sensitivity of the amplitude of the observation well responses with regard to specific storage, next to the recorded pressure response curves, type curves as a function of specific storage (the specific storage values differ by two orders of magnitude) are illustrated in Fig. 10b.

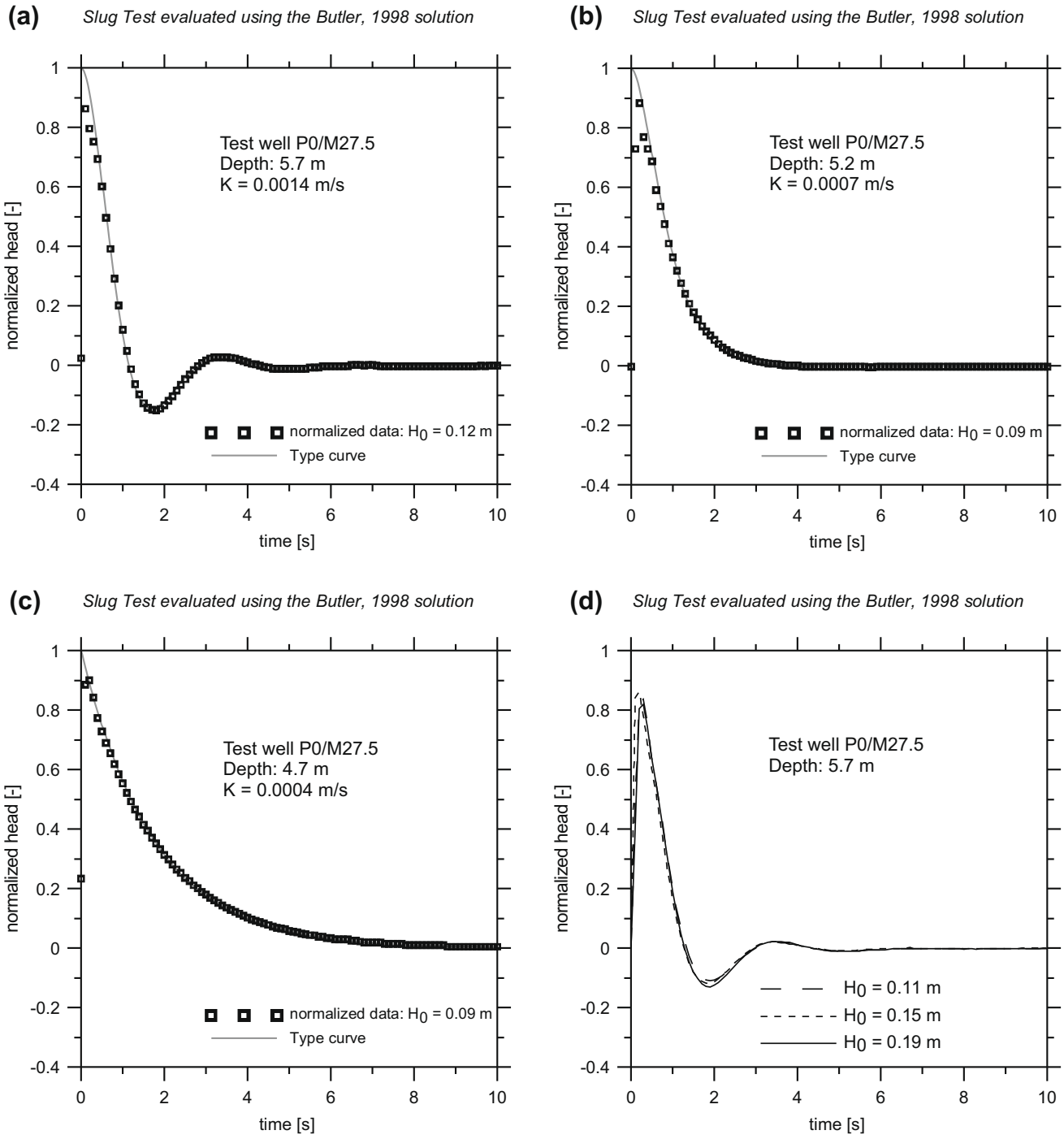


Fig. 7. Normalized head versus time plots. (a–c) Hydraulic conductivity estimates based on single-well slug tests for three chosen configurations. (d) Illustration of the influence of the initial displacement on the response curves.

The investigation of other test configurations, within the five-point star configuration, has shown that (a) the uncertainty of specific storage estimates never exceed the factor of two and that the hydraulic conductivity estimates are not significantly influenced by nonlinear effects. Note, a deviation of factor of two in specific storage estimates is acceptable for the most field applications.

Table 1 lists K and S_s estimates obtained for test configurations, where the double packer systems are positioned at the same depth. This test configuration allows the identification of potential vertical variations of hydraulic properties of the cross-well depth intervals. The hydraulic data (K and S_s) between the center well P0/M25 and well P0/M22.5 show a relatively uniform parameter distribution with depth. This is in agreement with the results from single well multi-level slug tests, which also show no significant depth variation. The only variation can be observed in the S_s values which increase by half an order of magnitude close to the aquifer top. In contrast, the measurements between the center well P0/M25 and the other three wells P0/M27.5, P2.5/M25, and PM2.5/M25 display significant variations with depth. The area close to the top of the aquifer is characterized by lower K values of 5×10^{-4} m/s and higher S_s values of 10^{-3} m^{-1}. With increasing depth the hydraulic conductivity increases and reaches 10^{-3} m/s close to the aquifer

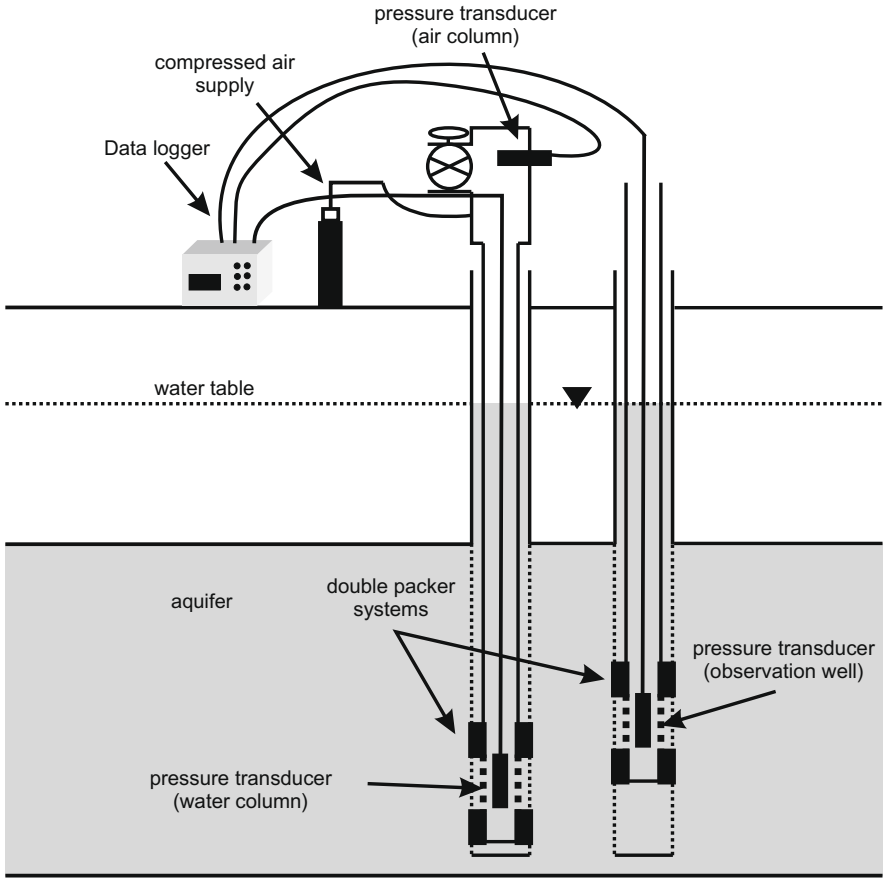


Fig. 8. Sketch of the test set-up used for the cross-well slug interference tests.

bottom. The specific storage values show an opposite behavior, i.e. they decrease to values of approx. 5×10^{-5} m^{-1} close to the bottom of the aquifer.

Travel time inversion approach

Although the presented evaluation of cross-well slug interference tests based on analytical solutions already provides detailed information about the hydraulic subsurface properties, it is not possible to assess hydraulic strata connectivity, which is important in identifying preferential pathways or barriers to groundwater flow and transport. This arises from the fact that the applied analytical solutions are based on an assumption of homogeneous parameter distribution within the subsurface. To address this problem, a hydraulic travel time based inversion approach was applied to the cross-well response data. The key element of the approach is the following line integral relating the arrival time of a "hydraulic signal" to the reciprocal value of diffusivity (Vasco et al., 2000; Kulkarni et al., 2000),

$$\sqrt{t_{peak}(x_2)} = \frac{1}{\sqrt{6}} \int_{x_1}^{x_2} \frac{ds}{\sqrt{D(s)}} \tag{1}$$

where t_{peak} is the travel time of the peak of a signal from the point x_1 (source) to the observation point x_2 (receiver) along the propagation path (s) and D is the diffusivity.

Brauchler et al. (2003) describe a further development of the line integral enabling the inversion of additional travel time diagnostics besides the peak time of a Dirac pulse at the origin. A travel time diagnostic is defined as the time of occurrence of a certain feature of the transient pressure pulse. For example, the 10% travel time diagnostic ($t_{10\%}$ diagnostic) is the time at which the pressure pulse rises to 10% of its ultimate peak value. In Fig. 11 the $t_{10\%}$ diagnostic and the peak time are illustrated.

The propagation of a seismic signal can be described similarly by a line integral relating the travel time t, recorded between a source and a receiver, with the spatial velocity distribution v of an investigated area:

$$t = \int_{x_1}^{x_2} \frac{ds}{v(s)} \tag{2}$$

The inverse of velocity, $1/v(s)$ is termed the slowness. In our approach we utilize the similarity between Eqs. (1) and (2). This similarity allows the use of algorithms originally developed for seismic tomography for the inversion of transient pressure responses. The hydraulic inversion was performed with the commercial software package GeoTom 3D (Jackson and Tweeton, 1996), which is based on the Simultaneous Iterative Reconstruction Technique (SIRT) algorithm (Gilbert, 1972). For a detailed derivation of the SIRT algorithm with regard to the inversion of transient pressure responses we refer to Brauchler et al. (2007).

Usually, in seismic or radar tomography, the velocity contrasts do not exceed 30%. In hydraulic tomography, however, diffusivity contrasts of several orders of magnitude are common. To account for this considerably larger contrast, seismic ray tracing techniques are applied, which lead to more reliable reconstructions for large diffusivity contrasts (e.g. Peterson et al., 1985) in comparison to straight ray approaches. Note, the calculated trajectories cannot be treated as streamlines since they depend on the chosen travel time diagnostics (Brauchler et al., 2007; Cheng et al., 2009). The trajectories describe the propagation of pressure levels, following

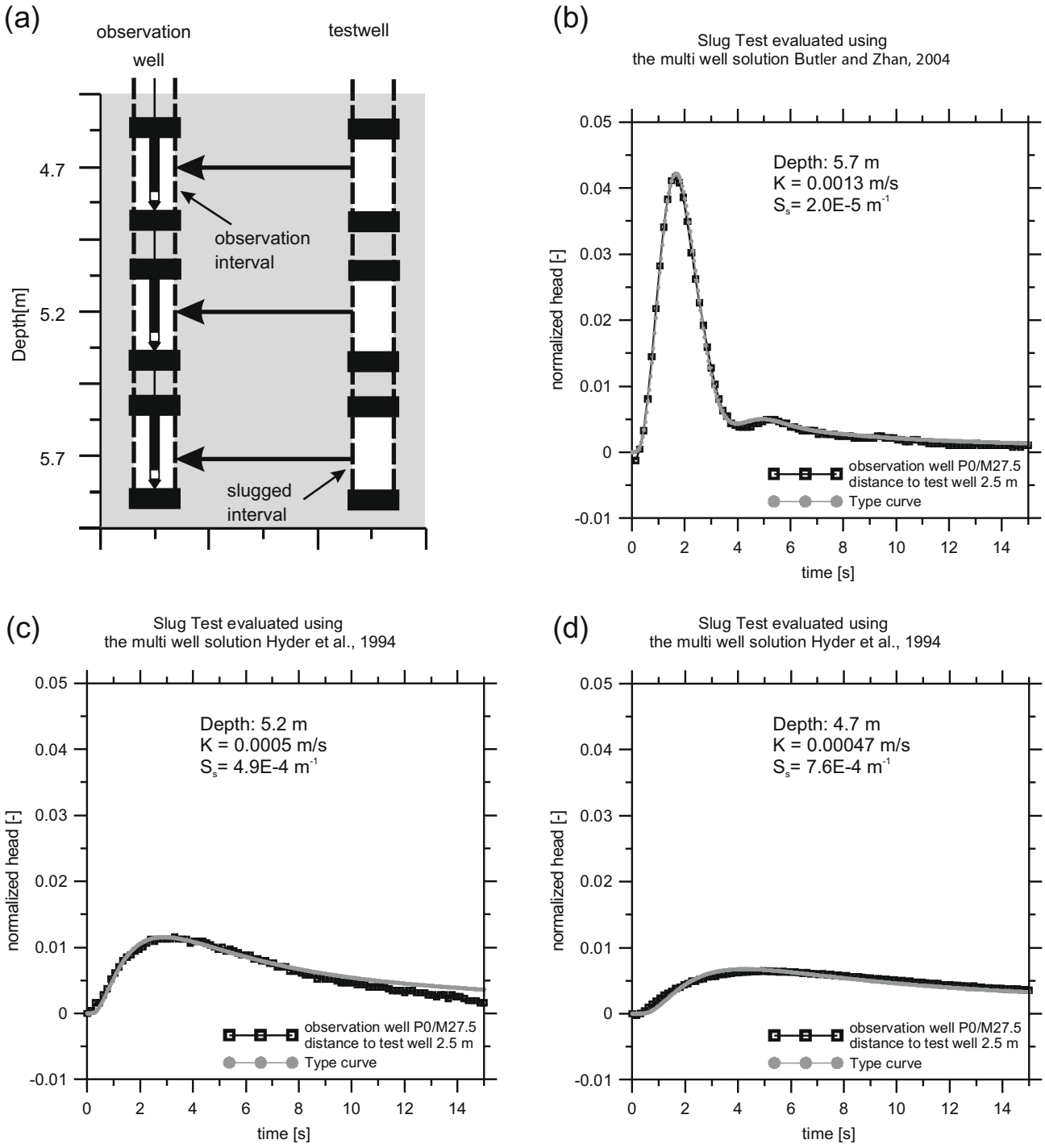


Fig. 9. Hydraulic conductivity and specific storage estimates based on cross-well slug interference tests for three chosen configurations.

Fermat's principle. Consequently, small travel time diagnostics, characterizing the initial part of a signal, follow the fastest pathways between source and receiver. The fastest pathways are usually preferential flow paths (if available). In contrast, late travel times, characterizing the final part of a signal, display integral behavior more or less, since the pressure difference between source and receiver extends across the entire system. Brauchler et al. (2007) verified this behavior by a numerical case study based on the simulation of cross-well slug interference tests arranged in a tomographic array. Therefore the travel time diagnostic of 10% was chosen, which is a compromise between obtaining high data quality (avoiding early time noise) and the findings of Brauchler et al. (2007), that early travel time diagnostics are more suited to resolve hydraulic heterogeneity. A more explicit discussion of the series expansion methods and different ray tracing algorithms are beyond the scope of this paper. Interested readers are referred to Jackson and Tweeton (1996), Um and Thurber (1987), Dines and Lytle (1979), Gilbert (1972) and Lo and Inderwiesen (1957).

Influence of wellbore storage and inertial effects on pressure pulse propagation

The impact of wellbore storage S_c, which is defined as the relative volume change ΔV per unit head change Δh in the well bore

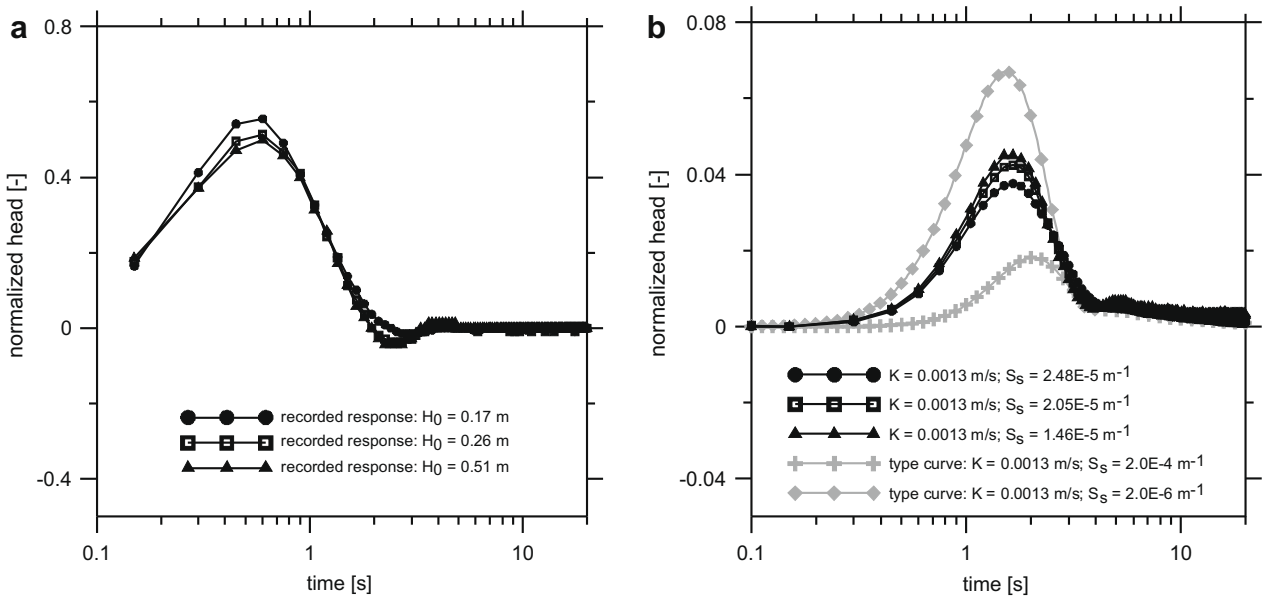


Fig. 10. Illustration of the influence of the initial displacement on the response curves. (a) Test well and (b) observation well.

Table 1
Hydraulic conductivity and specific storage estimates derived from type curve analysis of cross-well slug interference tests. Depth of stress and observation well test interval is related to middle of the double packer system.

Depth of stress and observation well test interval [m]	P0/M25–P0/M22.5		P0/M25–P0/M27.5		P0/M25–P2.5/M25		P0/M25–PM2.5/M25		Arithmetic mean	
	K [m/s]	S_s [m^{-1}]	K [m/s]	S_s [m^{-1}]	K [m/s]	S_s [m^{-1}]	K [m/s]	S_s [m^{-1}]	K [m/s]	S_s [m^{-1}]
5.71	4.0E−4	6.7E−	1.3E−3	2.0E−5	1.3E−3	1.7E−5	1.1E−3	7.8E−5	1.0E−3	1.9E−4
5.46	4.6E−4	4.5E−	1.1E−3	8.1E−5	6.2E−4	2.0E−4	2.2E−3	1.2E−4	1.1E−3	2.1E−4
5.21	5.0E−4	4.9E−	5.0E−4	4.9E−4	9.1E−4	6.6E−4	4.4E−3	2.9E−4	1.6E−3	4.8E−4
4.96	5.0E−4	4.6E−	3.0E−4	8.9E−4	7.7E−4	1.0E−3	5.8E−4	5.9E−4	5.4E−4	7.4E−4
4.71	4.1E−4	5.0E−	4.7E−4	7.6E−4	5.6E−4	9.2E−4	5.5E−4	7.9E−4	4.9E−4	7.4E−4
4.46	5.1E−4	1.2E−	3.3E−4	9.9E−4	–	–	6.5E−4	1.1E−3	4.9E−4	1.1E−3
4.21	3.2E−4	1.4E−	2.6E−4	1.6E−3	–	–	7.0E−4	1.3E−3	4.2E−4	1.4E−3

($S_c = \Delta V/V\Delta h$), on the propagation of pressure pulses is a key factor in the interpretation of inversion results related to typical field conditions. Wellbore storage affects the recorded signal of a pulse interference test in two ways. Firstly, for a given distance between the pulsed and the observation well, wellbore storage causes a delay in the time at which the pressure change is observed. Secondly, the response amplitude grows with the wellbore storage assuming that the head change Δh remains the same (Prats and Scott, 1975). In the presented field examples the influence of wellbore storage is sufficiently high, that the resulting time delay has to be corrected to properly invert the data.

For the correction of the travel times influenced by wellbore storage the procedure proposed by Brauchler et al. (2007) was applied. The peak arrival time of the pressure change was calculated in two different ways by: (a) neglecting the influence of the wellbore storage and (b) taking wellbore storage into account. (a) The peak time t_{peak} can be expressed as a function of the hydraulic properties S_s and K and the distance r as follows $t_{peak} = \frac{S_s r^2}{6K}$. For the derivation of this equation we refer to Brauchler et al. (2003). (b) For the calculation of the peak arrival time influenced by wellbore storage the analytical solution developed by Hyder et al. (1994) was applied. This solution is perfectly suited because it incorporates the effects of partial penetration and wellbore storage.

Consequently, a conversion factor can be defined, which corrects the measurements for the effects of wellbore storage as follows:

$$C_j = \frac{t_j^{(a)}}{t_j^{(b)}} \tag{3}$$

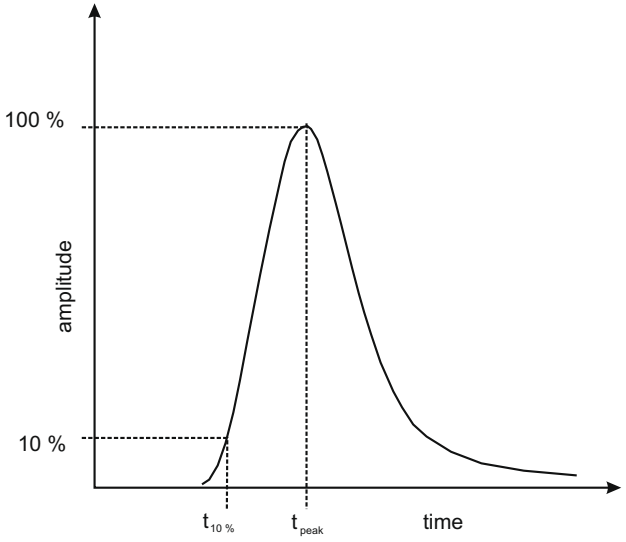


Fig. 11. Illustration of the 10% travel time diagnostic used in this work.

where C_j is the conversion factor of the jth travel time $t_j^{(a)}$ based on the relationship $t_{peak} = \frac{S_s r^2}{6K}$ and $t_j^{(b)}$ is the jth travel time based on the analytical solution developed by Hyder et al. (1994). For the correction of the travel times which are affected additionally by inertial mechanisms the same procedure was employed, with the exception that the analytical solution developed by Butler and Zhan (2004), accounting for inertial effects and wellbore storage, was used instead of the solution developed by Hyder et al. (1994).

In the presented field application example, the travel time delay of the 10% diagnostic caused by wellbore storage amounts on average to 40%. For the data which are additionally influenced by inertial effects the travel time delay is even 440%.

Inversion results

Four reconstructed diffusivity profiles (in the following termed tomograms), inverted using the procedure described above, are displayed in Fig. 12. All tomograms have in common that they were inverted separately and that the center well P0/M25 was used as a test well for all measured configurations. In Fig. 12a the West–East-section consisting of the tomograms P0/M25–P0/M22.5 and P0/M25–P0/M27.5 and in Fig. 12b the North–South-section consisting of the tomograms P0/M25–P2.5/M25 and P0/M25–PM2.5/M25 are illustrated. It is apparent that the tomograms in West–East direction (Fig. 12a) generally match at their interface (center well P0/M25) as well as the two tomograms in North–South-direction (Fig. 12b). This suggests that the inversion approach can successfully be applied to field data and that the experimental design for the performance of cross-well slug testing provides sufficient data accuracy with regard to the temporal resolution. The illustrated tomograms show a hydraulic diffusivity range between 1 and 10 m^2/s. The range encompasses the arithmetic means of hydraulic properties estimated by type curve evaluation of the 196 performed cross-well slug interference tests: arithmetic mean for hydraulic diffusivity: 2.63 m^2/s (arithmetic mean for hydraulic conductivity: 7.8×10^{-4} m/s, arithmetic mean for specific storage: $8.2 \times 10^{-4}\ m^{-1}$.

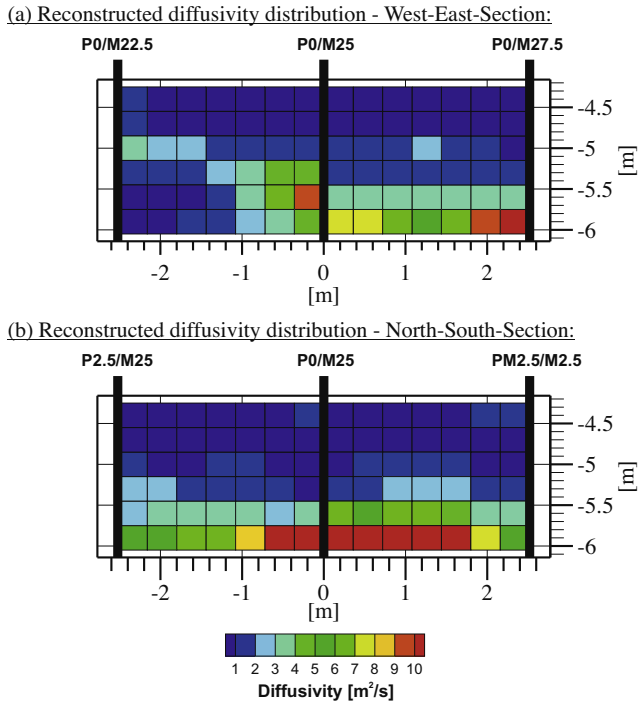


Fig. 12. Reconstructed diffusivity tomograms. Note each of the four illustrated tomograms is inverted separately.

Note, mean diffusivity (D_m) was calculated applying the following equation: $D_m = \sum_{i=1}^{196} \frac{k_i}{S_s} \cdot \frac{1}{196}$.

All tomograms show except for the left part of tomogram P0/M25–P0/M22.5, that the highest hydraulic diffusivity values are measured close to the bottom of the aquifer. This agrees with the hydraulic conductivity value estimated by multi-level single-well slug tests (Fig. 6). The vertical variations determined by means of type curve evaluation of the cross-well slug interference tests also follow the diffusivity distribution indicated by the tomograms. Generally, the highest K values and lowest S_s values are found at the lower part of the aquifer while the upper part of the aquifer is characterized by lower K and higher S_s estimates.

Tomogram P0/M25–P0/M22.5 demonstrates the potential of the travel time based tomographic approach for the identification of lateral changes. On the right hand side of the tomogram the high diffusivity zone at the aquifer bottom can be still recognized while the left part of the tomogram, in the vicinity of well P0/M22.5, is characterized by a more or less homogeneous diffusivity distribution. This indicates that the high diffusivity zone at the bottom of the aquifer pinches out towards well P0/M22.5.

The existence of the lateral change between well P0/M25 and P0/M22.5 can be also proven by multi-level single-well slug tests, conducted in the two wells. The multi-level single-well slug tests performed in well P0/M25 show the vertical K-distribution characteristics for the investigated area with higher values close to the bottom and the low values at the top of the aquifer. However the K-distribution in well P0/M22.5 is characterized by a more or less homogeneous K-distribution (Fig. 6). The best explanation for the differences in the K-distributions of the two wells is the existence of a lateral change in hydraulic properties near the base of the aquifer between the two wells.

The agreement between the hydraulic diffusivity tomograms and the results of the multi-level single well and cross-well slug interference tests based on type curve analysis demonstrates the reliability and significance of the travel time based inversion approach to depict lateral and vertical changes in hydraulic diffusivity conditions.

The West–East- and North–South-sections are shown together in Fig. 13. It is apparent that the two sections do not match in detail in the vicinity of the center well P0/M25. This result is surprising since the two profiles of the West–East- and the two profiles of the North–South-section match well at their individual interfaces. A possible explanation for this mismatch at the interface between the two sections could be hydraulic anisotropy within sedimentary architectural elements due to complex sedimentation processes.

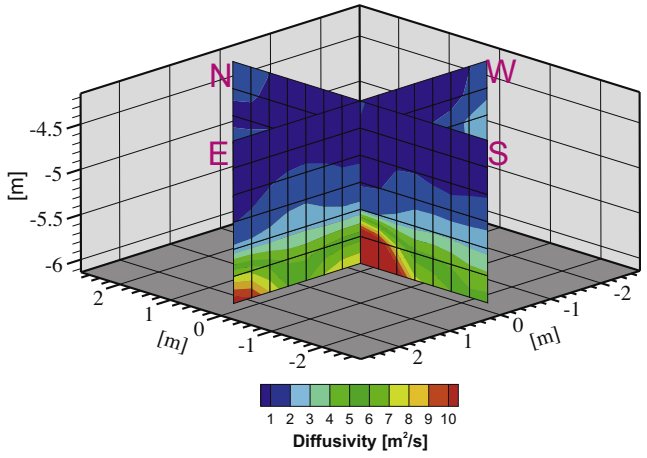


Fig. 13. Three-dimensional arrangement of the reconstructed diffusivity tomograms. Note each of the four illustrated tomograms is inverted separately.

However, the trend from lower diffusivities at the aquifer top to higher diffusivities at the aquifer bottom is still observed for both sections.

Summary and conclusions

The potential of cross-well slug interference tests for a high resolution characterization of aquifer heterogeneity was investigated at an experimental research site with a closely spaced network of 16 groundwater wells. The aquifer top and the bottom boundaries were assessed using geophysical well logs. To assess lateral changes of aquifer material distribution a seismic refraction survey was conducted. The seismic profiles indicate that the aquifer bottom is characterized by a strongly varying relief. The comparison of the seismic profiles with the geophysical well logs indicated that a simple reconstruction of the aquifer bottom by linear interpolation based on the geophysical well logs would yield an unacceptable simplification of the subsurface conditions.

Based on the previously collected detailed site characterization data an area, characterized by a constant aquifer thickness of 2 m and an average hydraulic conductivity of 5.0×10^{-4} m/s (from pumping tests), was selected for the field tests. Within this area five wells were installed, arranged as a five-point star with a distance of 2.5 m to the center well. For the determination of hydraulic conductivity multi-level single-well slug tests were performed in each well using a double packer system. The test results provide detailed information about vertical changes of hydraulic conductivity within the aquifer at each well location.

For obtaining a high resolution characterization of subsurface heterogeneity, four vertical profiles were recorded, each consisting of 49 cross-well slug interference tests. The transient pressure response curves were evaluated using type curve analysis and a travel time based tomographic inversion approach. The type curve analysis of the cross-well slug interference tests provided detailed information about the vertical changes of hydraulic conductivity and specific storage. The type curve evaluation of the multi-level single well and cross-well slug interference tests indicate potential lateral changes in interwell hydraulic properties. The spatial position of the lateral changes in hydraulic properties could be reconstructed by a travel time based tomographic inversion approach. The approach was chosen because of its high structural resolution and its computational effectiveness. It was possible to invert travel times of 49 response curves simultaneously on a PC within a few seconds. The tomographic inversion results were corrected to wellbore storage and inertial effects using an analytical model method reported in Brauchler et al., 2007. The analytical solution approach allows to generate a reference model for the quantification of the associated delays. This is an important result because the quantification of the delays by means of a numerical flow model would be very complex and would increase computational effort.

The combined evaluation of cross-well slug interference tests based on type curve analysis and travel time inversion could become a valuable tool for long term monitoring and the improvement of well established site characterization procedures. For example, cross-well slug interference tests performed in a tomographic array could allow detailed monitoring of clogging processes in an aquifer during artificial recharge (Pavelic et al., 2006), or the monitoring of hydraulic properties of permeable reactive barriers, which may change due to chemical processes (Morrison et al., 2006; Henderson and Demond, 2007). The approach can be also applied in cases where a dense well network is available, e.g. at control planes used for contaminant site characterization methods such as integral pumping tests (e.g. Ptak et al., 2000; Herold et al., 2009).

Acknowledgements

The investigations were conducted with the financial support of the German Research Foundation within the project "High resolution aquifer characterization based on direct-push technology: An integrated approach coupling hydraulic and seismic tomography" (Grant No. BR3379/1-2). Infrastructure and logistic support were provided by the Stadtwerke Göttingen AG. This manuscript greatly benefited from comments and suggestions by J.J. Butler Jr. and the three reviewers for Journal of Hydrology.

References

Audouin, O., Bodin, J., 2007. Cross-borehole slug test analysis in a fractured limestone aquifer. J. Hydrol.. doi:10.1016/j.jhydrol.2007.10.021.
Barker, J., 1988. A generalized radial flow model for hydraulic tests in fractured rock. Water Resour. Res. 24 (10), 1796–1804.
Barker, J.A., Black, J.H., 1983. Slug tests in fissured aquifers. Water Resour. Res. 19 (6), 1558–1564.
Belitz, K., Dripps, W., 1999. Cross-well slug testing in unconfined aquifers: a case study from the sleepers river watershed, Vermont. Ground Water 37 (3), 438–447.
Brauchler, R., Liedl, R., Dietrich, P., 2003. A travel time based hydraulic tomographic approach. Water Resour. Res. 39 (12), 1370. doi:10.1029/2003WR002262.
Brauchler, R., Cheng, J., Everette, M., Johnson, B., Dietrich, P., Liedl, R., Sauter, M., 2007. An integrated approach coupling hydraulic travel time tomography and amplitude inversion. J. Hydrol.. doi:10.1016/j.jhydrol.2007.08.011.
Butler Jr., J.J., 1995. Slug tests with observation wells; extension of Hyder et al. (1994) Solution to case of head in an observation well. Kansas Geological Survey, Open-file Report no. 95–43, 14 pp.
Butler Jr., J.J., 1998. The Design, Performance, and Analysis of Slug Tests. Lewis Pub.. 252 pp.
Butler Jr., J.J., Zhan, X., 2004. Hydraulic tests in highly permeable aquifers. Water Resour. Res. 40, W12402. doi:10.1029/2003WR002998.
Butler Jr., J.J., Garnett, E.J., Healey, J.M., 2003. Analysis of slug tests in formations of high hydraulic conductivity. Ground Water 41 (5), 620–630.
Cheng, J., Brauchler, R., Everett, M., 2009. Comparison of early and late travel times of pressure pulses induced by multilevel slug tests. Eng. Appl. Comput. Fluid Mech. 3 (4), 529–541.
Christy, C.D., Christy T.M., Wittig, V., 1994. A percussion probing tool for the direct sensing of soil conductivity. In: Proceedings of the 8th National Outdoor Action Conference, Westerville. National Ground Water Association, Ohio, pp. 381–394.
Chu, L., Grader, A.S., 1991. Transient pressure analysis for an interference slug test. SPE Paper 23444, Society of Petroleum Engineering.
Datta-Gupta, A., Kulkarni, K.N., Yoon, S., Vasco, D.W., 2001. Streamlines ray tracing and production tomography: generalization to compressible flow. Petrol. Geosci. 7, 75–86.
Dines, K.A., Lytle, R.J., 1979. Computerized geophysical tomography. Proc. IEEE 67 (7), 1065–1073.
Ferris, J.G., Knowles, D.B., 1963. The slug-injection test for estimating the coefficient of transmissibility of an aquifer. In: US Geological Survey Water-Supply Paper 15361-I, R. Bentall (compiler), p. 299.
Gilbert, P., 1972. Iterative methods for three-dimensional reconstruction of an object from projections. J. Theor. Biol. 36, 105–117.
He, Z., Datta-Gupta, A., Vasco, D.W., 2006. Rapid inverse modeling of pressure interference tests using trajectory-based travel time and amplitude sensitivities. Water Resour. Res. 42, W03419. doi:10.1029/2004WR003783.
Henderson, A.D., Demond, A.H., 2007. Long-term performance of zero-valent iron permeable reactive barriers: a critical review. Environ. Eng. Sci.. doi:10.1089/ees.2006.0071.
Herold, M., Ptak, T., Bayer-Raich, M., Wendel, T., Grathwohl, P., 2009. Integral quantification of contaminant mass flow rates in a contaminated aquifer: conditioning of the numerical inversion of concentration-time series. J. Contam. Hydrol. 106 (1–2), 29–38.
Huggenberger, P., Regli, C., 2006. A sedimentological model to characterize braided river deposits for hydrogeological applications. In: Sambrook-Smith, G., Best, J., Bristow, C., Petts, G. (Eds.), Braided Rivers, Process, Deposits, Ecology and Management, vol. 36. ISA Special Publication, pp. 51–74.
Hyder, Z., Butler Jr., J.J., McElwee, C.D., Liu, W., 1994. Slug test in partially penetrating wells. Water Resour. Res. 30 (8), 2945–2957.
Jackson, M.J., Tweeton, D.R., 1996. 3DTOM: Three-dimensional Geophysical Tomography. United States Department of the Interior, Bureau of Mines, Report of Investigation 9617, 84 pp.
Karasaki, K., Long, J., Witherspoon, P., 1988. Analytical models of slug tests. Water Resour. Res. 24 (1), 115–126.
Kulkarni, K.N., Datta-Gupta, A., Vasco, D.W., 2000. A streamline approach to integrating transient pressure data into high resolution reservoir models. SPE J. 6 (3).
Lo, T., Inderwiesen, P.L., 1957. Fundamentals of seismic tomography. Geophysical monograph series (6), 178 pp.

Morrison, S.J., Mushovic, P., Niesen, P.L., Preston, L., 2006. Early breakthrough of molybdenum and uranium in a permeable reactive barrier. Environ. Sci. Technol. 40, 2018–2024.

Novakowski, K., 1989. Analysis of pulse interference tests. Water Resour. Res. 25 (11), 2377–2387.

Pavelic, P., Dillon, P.J., Karen, E.B., Gerges, N.Z., 2006. Hydraulic evaluation of aquifer storage and recovery (ASR) with urban stormwater in a brackish limestone aquifer. Hydrol. J.. doi:10.1007/s10040-006-0078-4.

Peterson, J.E., Paulsson, B.N., McEvilly, T.V., 1985. Application of algebraic reconstruction technique to crosshole seismic data. Geophysics 50, 1566–1580.

Prats, M., Scott, J.B., 1975. Effect of wellbore storage on pulse-test pressure response. J. Petrol. Technol., 707–709.

Ptak, T., Schwarz, R., Holder, T., Teutsch, G., 2000. Ein neues verfahren zur quantifizierung der grundwasseremission, teil 2: numerische Lösung und anwendung in eppelheim. Grundwasser 4, 176–183.

Rovey, C.W., Cherkauer II, D.S., 1995. Scale dependency of hydraulic conductivity measurements. Ground Water 33 (5), 769–780.

Sageev, A., 1986. Slug test analysis. Water Resour. Res. 22 (8), 1323–1333.

Schulmeister, M.K., Butler Jr., J.J., Healey, J.M., Zheng, L., Wysocki, D.A., McCall, W., 2003. Direct-push electrical conductivity logging for high-resolution hydrostratigraphic characterization. Ground Water Monit. Rem. 23 (3), 52–62.

Sheriff, R.E., Geldart, L.P., 1995. Exploration Seismology. Cambridge Univ. Press. 592 pp.

Spane Jr., F.A., 1996. Applicability of slug interference tests for hydraulic characterization of unconfined aquifers: (1) analytical assessment. Ground Water 34 (1), 66–74.

Spane Jr., F.A., Thorne, P.D., Swanson, L.C., 1996. Applicability of slug interference tests for hydraulic characterization of unconfined aquifer: (2) field test examples. Ground Water 34 (5), 925–933.

Um, J., Thurber, C., 1987. A fast algorithm for two-point seismic ray tracing. Bull. Seismol. Soc. Am. 77 (3), 972–986.

Vasco, D.W., Datta-Gupta, A., 1999. Asymptotic solutions for solute transport: a formalism for tracer tomography. Water Resour. Res. 35 (1), 1–16.

Vasco, D.W., Karasaki, K., 2006. Interpretation and inversion of low-frequency head observations. Water Resour. Res. 42, W05408. doi:10.1029/2005WR004445.

Vasco, D.W., Yoon, S., Datta-Gupta, A., 1999. Integrating dynamic data into high resolution reservoir models using streamline based analytic sensitivity coefficients. SPEJ Soc. Petrol. Eng. J. 4, 1–11.

Vasco, D.W., Keers, H., Karasaki, K., 2000. Estimation of reservoir properties using transient pressure data: an asymptotic approach. Water Resour. Res. 36 (9), 3447–3465.

Virieux, J., Flores-Luna, C., Gibert, D., 1994. Asymptotic theory for diffusive electromagnetic imaging. Geophys. J. Int. 119, 857–868.